CBSE Class 10 Mathematics - Test Paper — Solutions

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1. **1.** The decimal expansion of $\frac{147}{1200}$ will terminate after how many places of decimal?

   Answer:

   $$(C)4$$

   Solution:

   1. Step 1: Simplify the fraction $\frac{147}{1200}$ by dividing numerator and denominator by 3 to get $\frac{49}{400}$.
   2. Step 2: Factorise the denominator: $400 = 2^4 \times 5^2$.
   3. Step 3: For a fraction in simplest form, the decimal expansion terminates if the denominator is of the form $2^m \times 5^n$.
   4. Step 4: The highest power among 2 and 5 in the denominator is $\max(4, 2) = 4$, so the decimal expansion terminates after 4 decimal places.

   Derivation:

   $$\frac{147}{1200} = \frac{49}{400} = \frac{49}{2^4 \times 5^2} Since denominator is of the form 2^m \times 5^n, decimal expansion terminates after r$$

   Real Numbers — Terminating and Non-terminating Decimals

2. **2.** If $A = 2^3 \times 5^2$ and $B = 2^2 \times 5^3 \times 7$, find the HCF of A and B.

   Answer:

   $$2^2 \times 5^2 = 100$$

   Solution:

   1. Write the prime factorisations of A and B.
   2. Identify common prime factors with the smallest exponent.
   3. Multiply the common prime factors with their smallest exponents.

   Derivation:

   $$\text{HCF} = 2^{\min(3,2)} \times 5^{\min(2,3)} \times 7^0 = 2^2 \times 5^2 = 4 \times 25 = 100$$

   Real Numbers — HCF and LCM using prime factorisation

3. **3.** Divide the polynomial $p(x) = x^3 - 3x^2 + 5x - 3$ by $g(x) = x^2 - 2$ using the division algorithm. Find the quotient $q(x)$ and remainder $r(x)$.

   Answer:

   $$Quotient : x - 3, Remainder : 7x - 9$$

   Solution:

1. Step 1: Divide the leading term of $p(x)$ by the leading term of $g(x)$: $x^3 \div x^2 = x$. Multiply $g(x)$ by x: $x(x^2 - 2) = x^3 - 2x$. Subtract from $p(x)$ to get the first remainder: $(x^3 - 3x^2 + 5x - 3) - (x^3 - 2x) = -3x^2 + 7x - 3$.
2. Step 2: Divide the leading term of the new polynomial by the leading term of $g(x)$: $-3x^2 \div x^2 = -3$. Multiply $g(x)$ by -3: $-3(x^2 - 2) = -3x^2 + 6$. Subtract: $(-3x^2 + 7x - 3) - (-3x^2 + 6) = 7x - 9$. Since the degree of $7x - 9$ is less than the degree of $g(x)$, this is the remainder. The quotient is $x - 3$.

Derivation:

$$\begin{array}{r|l} x^3 - 3x^2 + 5x - 3 & x^2 - 2 \\ \hline x^3 - 2x & x - 3 \\ \hline -3x^2 + 7x - 3 & \\ -3x^2 + 6 & \\ \hline 7x - 9 & \end{array}$$

Polynomials — division algorithm for polynomials

4. **4.** Solve the following pair of equations by reducing them to a pair of linear equations: $\frac{2}{x} + \frac{3}{y} = 11$ and $\frac{5}{x} - \frac{4}{y} = -7$

Answer:

$$x = 1, y = 1$$

Solution:

1. Let $p = \frac{1}{x}$ and $q = \frac{1}{y}$ to convert the equations into linear form.
2. Substitute to get: $2p + 3q = 11$ and $5p - 4q = -7$.
3. Solve the linear equations using elimination: Multiply first equation by 4 and second by 3, then add to eliminate q: $8p + 12q = 44$ and $15p - 12q = -21$ =¿ $23p = 23$ =¿ $p = 1$.
4. Substitute $p = 1$ into $2p + 3q = 11$: $2(1) + 3q = 11$ =¿ $3q = 9$ =¿ $q = 3$. Thus $x = \frac{1}{p} = 1$ and $y = \frac{1}{q} = \frac{1}{3}$.

Derivation: Given equations: $\frac{2}{x} + \frac{3}{y} = 11$ ...(i) $\frac{5}{x} - \frac{4}{y} = -7$ ...(ii) Let $p = \frac{1}{x}$, $q = \frac{1}{y}$. Then (i) becomes $2p + 3q = 11$, (ii) becomes $5p - 4q = -7$. Multiply (i) by 4: $8p + 12q = 44$ Multiply (ii) by 3: $15p - 12q = -21$ Adding: $23p = 23 \implies p = 1$ Substitute $p = 1$ in $2p + 3q = 11$: $2 + 3q = 11 \implies 3q = 9 \implies q = 3$ So $x = \frac{1}{p} = 1$, $y = \frac{1}{q} = \frac{1}{3}$. *Pair of Linear Equations in Two Variables — equations reducible to linear form*

5. **5.** A train travels a distance of 480 km at a uniform speed. If the speed had been 8 km/h less, then it would have taken 3 hours more to cover the same distance. Find the usual speed of the train. (Note: Form a quadratic equation and solve it.)

Answer:

$$Usualspeedofthetrain = 40km/h$$

Solution:

1. Let the usual speed of the train be x km/h.
2. Time taken at usual speed = distance/speed = $\frac{480}{x}$ hours.

3. If speed is 8 km/h less, new speed = $(x-8)$ km/h.
4. Time taken at reduced speed = $\frac{480}{x-8}$ hours.
5. According to the problem, the time at reduced speed is 3 hours more than the usual time: $\frac{480}{x-8} = \frac{480}{x} + 3$.
6. Multiply both sides by $x(x-8)$ to clear denominators: $480x = 480(x-8) + 3x(x-8)$.
7. Simplify: $480x = 480x - 3840 + 3x^2 - 24x$.
8. Cancel $480x$ on both sides: $0 = -3840 + 3x^2 - 24x$.
9. Rearrange: $3x^2 - 24x - 3840 = 0$. Divide through by 3: $x^2 - 8x - 1280 = 0$.
10. Solve the quadratic equation: $x^2 - 8x - 1280 = 0$. Using quadratic formula $x = \frac{-b\pm\sqrt{b^2-4ac}}{2a}$ with $a = 1$, $b = -8$, $c = -1280$.
11. Discriminant $D = (-8)^2 - 4(1)(-1280) = 64 + 5120 = 5184$.
12. So $x = \frac{8\pm\sqrt{5184}}{2} = \frac{8\pm72}{2}$.
13. Thus $x = \frac{8+72}{2} = 40$ or $x = \frac{8-72}{2} = -32$.
14. Since speed cannot be negative, $x = 40$ km/h.

Derivation: Let usual speed = x km/h. Usual time = $\frac{480}{x}$ h. Reduced speed = $(x-8)$ km/h. New time = $\frac{480}{x-8}$ h. Condition: $\frac{480}{x-8} = \frac{480}{x} + 3$. Multiply by $x(x-8)$: $480x = 480(x-8) + 3x(x-8)$. Simplify: $480x = 480x - 3840 + 3x^2 - 24x$. Cancel $480x$: $0 = 3x^2 - 24x - 3840$. Divide by 3: $x^2 - 8x - 1280 = 0$. Quadratic formula: $x = \frac{8\pm\sqrt{64+5120}}{2} = \frac{8\pm72}{2}$. Positive root: $x = 40$. *Quadratic Equations — Word Problems (Speed-Distance-Time)*

6. **6.** In $\triangle ABC$, P and Q are points on sides AB and AC respectively such that $PQ \parallel BC$. The area of $\triangle APQ$ is 16 cm^2 and the area of trapezium $PBCQ$ is 20 cm^2. Find the ratio $AP : PB$. If $AB = 12$ cm, find the lengths of AP and PB.

Answer:

$$AP : PB = 2 : 1, AP = 8cm, PB = 4cm$$

Solution:

1. Let $AP = x$, $PB = y$, so $AB = x + y$.
2. Since $PQ \parallel BC$, $\triangle APQ \sim \triangle ABC$ (AA similarity).
3. Ratio of areas of similar triangles equals square of ratio of corresponding sides: $\frac{\text{area}(\triangle APQ)}{\text{area}(\triangle ABC)} = \left(\frac{AP}{AB}\right)^2$.
4. Area of $\triangle ABC = \text{area}(\triangle APQ) + \text{area}(PBCQ) = 16 + 20 = 36$ cm^2.
5. Thus $\frac{16}{36} = \left(\frac{x}{x+y}\right)^2 \Rightarrow \frac{4}{9} = \left(\frac{x}{x+y}\right)^2 \Rightarrow \frac{x}{x+y} = \frac{2}{3}$ (taking positive ratio).
6. Cross-multiply: $3x = 2x + 2y \Rightarrow x = 2y \Rightarrow \frac{x}{y} = \frac{2}{1}$. So $AP : PB = 2 : 1$.
7. Given $AB = 12$ cm, so $x + y = 12$. Since $x = 2y$, substitute: $2y + y = 12 \Rightarrow 3y = 12 \Rightarrow y = 4$ cm, then $x = 8$ cm.

Derivation:

$$\text{area}(\triangle ABC) = 16 + 20 = 36 \text{ cm}^2$$

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\begin{aligned}
\frac{\text{area}(\triangle APQ)}{\text{area}(\triangle ABC)} &= \left(\frac{AP}{AB}\right)^2 \\
\frac{16}{36} &= \left(\frac{x}{x+y}\right)^2 \\
\frac{4}{9} &= \left(\frac{x}{x+y}\right)^2 \\
\frac{x}{x+y} &= \frac{2}{3} \\
3x &= 2(x+y) \\
x &= 2y \\
AP : PB &= x : y = 2 : 1 \\
AB &= x + y = 12 \Rightarrow 2y + y = 12 \Rightarrow y = 4, x = 8
\end{aligned}
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Triangles — Similarity and Area Ratio